



$$\hat{\mathbf{s}}_{t+k} = P_\phi(\hat{\mathbf{s}}_{t+k-1}, \mathbf{a}_{t+k-1}), \quad \mathcal{L}_{multi} = \sum_{k=1}^K \lambda_k \|\hat{\mathbf{s}}_{t+k} - E_{\bar{\theta}}(o_{t+k})\|_F^2$$